So, in addition to the existing systems, the collection of irrational numbers and imaginary numbers (See Chapter 3) are to be adjoined. Consequently $\mathbb{R}$ and then $\mathbb{C}$ are obtained. The biggest number system $\mathbb{C}$ properly includes all the other number systems $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, and $\mathbb{R}$ as subsets.

Number System / Operation	$\mathbb{N}$	$\mathbb{Z}$	$\mathbb{Q}$	$\mathbb{R}$	$\mathbb{C}$	$\mathbb{Q}\setminus\{0\}$	$\mathbb{R}\setminus\{0\}$	$\mathbb{C}\setminus\{0\}$
$+$	Binary	Binary	Binary	Binary	Binary	Not Binary	Not Binary	Not Binary
$-$	Not Binary	Binary	Binary	Binary	Binary	Not Binary	Not Binary	Not Binary
$\times$	Binary	Binary	Binary	Binary	Binary	Binary	Binary	Binary
$\div$	Not Binary	Not Binary	Not Binary	Not Binary	Not Binary	Binary	Binary	Binary

Table12.1

Example12.1

Examine the binary operation (closure property) of the following operations on the respective sets (if it is not, make it binary):

(i) $a*b=a+3ab-5b^2;\forall a,b\in\mathbb{Z}$ (ii) $a*b=\left(\frac{a-1}{b-1}\right),\ \forall a,b\neq 1\in\mathbb{Q}$

Solution

(i) Since $\times$ is binary operation on $\mathbb{Z}$, $a,\ b\in\mathbb{Z}\Rightarrow a\times b=ab\in\mathbb{Z}$ and $b\times b=b^2\in\mathbb{Z}$ ... (1)

The fact that $+$ is binary operation on $\mathbb{Z}$ and (1) $\Rightarrow 3ab=(ab+ab+ab)\in\mathbb{Z}$ and $5b^2=(b^2+b^2+b^2+b^2+b^2)\in\mathbb{Z}$. .... (2)

Also $a\in\mathbb{Z}$ and $3ab\in\mathbb{Z}$ implies $a+3ab\in\mathbb{Z}$. ... (3)

(2), (3), the closure property of $-$ on $\mathbb{Z}$ yield $a*b=(a+3ab-5b^2)\in\mathbb{Z}$. Since $a*b$ belongs to $\mathbb{Z}$, * is a binary operation on $\mathbb{Z}$.

(ii) In this problem $a*b$ is in the quotient form. Since the division by 0 is undefined, the denominator $b-1$ must be nonzero.

It is clear that $b-1=0$ if $b=1$. As $1\in\mathbb{Q}$, $*$ is not a binary operation on the whole of $\mathbb{Q}$. However it can be found that by omitting 1 from $\mathbb{Q}$, the output $a*b$ exists in $\mathbb{Q}\setminus\{1\}$. Hence $*$ is a binary operation on $\mathbb{Q}\setminus\{1\}$.

12.2.2 Some more properties of a binary operation

Commutative property

Any binary operation $*$ defined on a nonempty set S is said to satisfy the commutative property, if

$$a*b=b*a\ \ \forall a,b\in S.$$

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